SMART HOME SHIELD: IOT DRIVEN REAL-TIME THREAT PROTECTION WITH PHOTO TO TELEGRAM APP

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MAJOR PROJECT STAGE - II REPORT

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CERTIFICATE

Date: */11/2025*

This is to certify that the project work entitled ***SMART HOME SHIELD: IOT DRIVEN REAL-TIME THREAT PROTECTION WITH PHOTO TO TELEGRAM APP*** *is a Bonafide work carried out by* ***K.MAHESH, K.SWETHA, K.PAVANI, B.PRESHITHA, V.ASHISH*** *bearing Roll Nos.* ***23RS5A0408, 23RS5A0406, 22RS1A0425, 22RS1A0406,*** *and* ***23RS5A0415*** *in the partial fulfilment of the requirements for the degree of* ***BACHELOR OF TECHNOLOGY*** *in* ***ELECTRONICS AND COMMUNICATION ENGINEERING*** *by the Jawaharlal Nehru Technological University Hyderabad during the academic year 2025-26.*

The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

The Smart Home Shield project is an innovative IoT-based real-time security system designed to protect homes from potential intrusions and unauthorized access. The system uses motion detection sensors and cameras to continuously monitor the surroundings. When any suspicious movement is detected, the camera instantly captures an image of the intruder and sends it directly to the user's Telegram app using an automated IoT communication module.

This immediate photo alert allows homeowners to take quick action even when they are away, ensuring round-the-clock safety. The system is built using components like Raspberry Pi (or ESP32), PIR motion sensor, and camera module, integrated with the Telegram API for instant notifications.

The Smart Home Shield combines artificial intelligence, cloud connectivity, and IoT technology to create a low-cost, efficient, and user-friendly security solution. It provides real-time monitoring, instant alerts, and evidence capture, making it a reliable system for modern smart homes.

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Chapter 1

Introduction

1.1 Introduction

Home security has become a major concern in modern society due to the increasing number of thefts, unauthorized entries, and safety threats. Traditional security systems like CCTV or alarm-based setups often require continuous monitoring and lack instant communication with the user when an intrusion occurs. With the advancement of IoT technologies, it is now possible to build automated, intelligent, and affordable security systems that work in real-time. This project, *Smart Home Shield*, uses IoT components to detect threats and instantly send alerts with photos to the user's Telegram app, providing a reliable and modern home protection solution.

Furthermore, smart security systems have become an essential part of modern homes because they offer continuous surveillance without requiring manual effort. The integration of IoT allows these systems to operate independently, analyse sensor data, and notify the user instantly. This ensures improved safety, reduced response time, and greater convenience compared to traditional monitoring methods.

1.2 Aim of the Project

The main aim of this project is to design and develop an IoT-based home security system that can detect intrusion, capture real-time images, and send them instantly to the user's Telegram application. The project focuses on creating a low-cost and efficient alternative to commercial surveillance systems. By integrating sensors like PIR, LDR, and ESP32-CAM, the system aims to provide continuous monitoring, immediate notifications, and enhanced safety for homes, offices, and restricted areas without requiring continuous human supervision.

1.3 Methodology

The project follows a systematic approach starting from component selection to final testing. First, suitable sensors such as PIR, LDR, and laser modules were chosen to detect different intrusion activities. Next, the ESP32-CAM module was programmed using Arduino IDE to capture images when a sensor triggers. A Telegram bot was created for sending real-time

notifications. The system was then connected to Wi-Fi, integrated with the bot API, and tested under different scenarios to ensure accurate detection and fast response.

1.4 Significance of the Work

This project holds significant importance in the field of affordable home automation and security. Many existing security solutions are either expensive or require professional installation and constant monitoring. The Smart Home Shield offers a simple, low-cost alternative that can be installed by anyone without technical difficulty. It provides instant alerts, live images, and real-time responses directly to the user's smartphone through Telegram, making it highly useful for homeowners, shop owners, and small businesses requiring an efficient and dependable security system.

1.5 Organization of the Thesis

This thesis is organized into several chapters for better understanding. Chapter 1 introduces the background, aim, methodology, and significance of the project. Chapter 2 gives a detailed overview of the ESP32 microcontroller and its features. Chapter 3 explains the hardware components used, along with their interfacing. Chapter 4 covers the software tools such as Arduino IDE and Telegram configuration. Finally, Chapter 5 presents the conclusions, results, and possible future enhancements.

Chapter 2

OVERVIEW OF ESP32

2.1 Introduction

The ESP32 is one of the most widely used microcontroller platforms in IoT applications due to its powerful dual-core processor, built-in Wi-Fi, Bluetooth capabilities, and extensive GPIO support. It offers high performance at a low cost, making it ideal for smart home automation, monitoring systems, wearable devices, and real-time data communication. Its flexibility, reliability, and low-power operation make it highly suitable for continuous security monitoring applications like the Smart Home Shield project.

2.2 Types of ESP Modules

The ESP family includes several modules such as the ESP8266, ESP01, ESP32 Devkit, ESP32-CAM, ESP32-S2, and ESP32-C3. Each module has unique features tailored for specific IoT applications. While ESP8266 is suitable for basic Wi-Fi tasks, ESP32 provides better processing power and more peripherals. ESP32-CAM is specially designed for image-based applications. Selecting the right module depends on requirements like camera support, GPIO availability, power consumption, and wireless connectivity.

1. ESP8266 Module

The ESP8266 is one of the earliest Wi-Fi-enabled microcontroller modules used in IoT applications. It features a single-core processor, built-in Wi-Fi, and supports basic digital and analog operations. Due to its low cost and easy programming using Arduino IDE, it became extremely popular for Wi-Fi automation. Although it lacks advanced features like Bluetooth or camera support, it is suitable for simple projects such as home automation, wireless sensor networks, and small IoT applications where low power and affordability are important.

2. ESP32 Devkit Module

The ESP32 Devkit is an upgraded version of the ESP8266, offering dual-core processing, integrated Wi-Fi and Bluetooth, more GPIO pins, and advanced peripherals. It supports ADC, DAC, PWM, UART, SPI, and I2C interfaces, making it versatile for complex IoT projects. Its improved performance enables multitasking, faster data handling, and low-power operations. This module is widely used in home automation, robotics, wearable devices, and real-time monitoring systems where stability and high processing speed are essential.

3. ESP32-CAM Module

The ESP32-CAM is a compact microcontroller board that includes the ESP32 chip along with an OV2640 camera module. It supports Wi-Fi, microSD storage, and onboard flash. This module is widely used in image-based IoT applications such as surveillance, face recognition, intrusion detection, and smart access systems. Its ability to capture and transmit photos in real time makes it ideal for security-related projects. Despite its small size, it delivers excellent performance and integrates easily with cloud services and smartphone apps.

2.3 ESP32

2.3.1 Features of ESP32

- Low power 32-bit CPU, can also serve the application processor.
- Up to 160MHz clock speed, summary computing power up to 600 DMIPS.
- Built-in 520 KB SRAM, external 4MPSRAM.
- Supports UART/SPI/I2C/PWM/ADC/DAC.
- Support OV2640 and OV7670 cameras, built-in flash lamp.
- Support image Wi-Fi upload.
- Supports multiple sleep modes.
- Embedded Lwip and Free RTOS.

- Supports STA/AP/STA+AP operation mode.
- Support Smart Config/AirKiss technology.
- Support for serial port local and remote firmware upgrades (FOTA).

2.3.2 Description of ESP32

The ESP32 microcontroller is built around the Xtensa LX6 architecture and is designed for low-power IoT applications. It integrates CPU cores, memory, wireless communication modules, and digital/analog interfaces into a single chip. The microcontroller can handle multiple tasks simultaneously, making it effective for applications requiring sensor monitoring, communication with cloud services, and real-time control. Its flexibility and modular design allow developers to create advanced smart systems.

2.3.3 Pin Description of ESP32

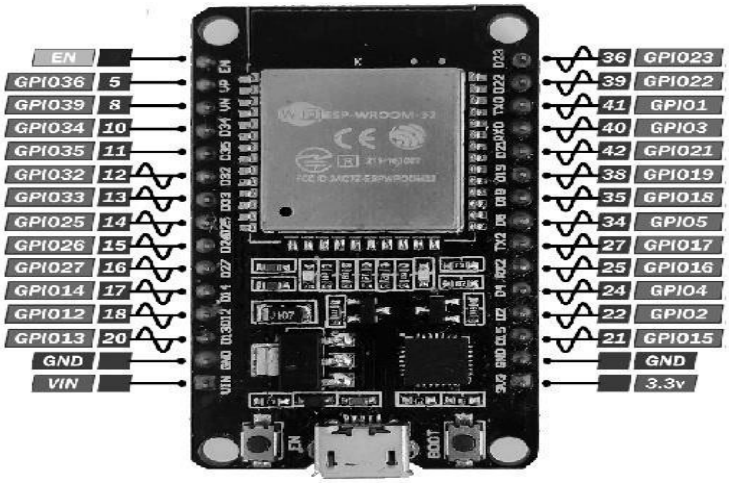

Fig 2.1: ESP32 Module

The ESP32 contains a wide variety of pins, including GPIO, ADC, DAC, UART, SPI, I2C, and PWM. These pins enable the connection of sensors, actuators, displays, and communication peripherals. The ADC pins allow analogy sensor measurements, while GPIO pins support digital input and output operations. Power pins provide options for 3.3V and ground connections. The versatile pin configuration allows the ESP32 to be used in complex embedded applications.

Additionally, the ESP32 pin structure is designed to support a wide variety of external modules, making it highly flexible for embedded applications. Some pins also have internal pull-up and pull-down resistors, allowing stable input readings. Special pins such as strapping pins control the boot mode, while high-current capable pins support driving LEDs, motors, and other output devices reliably without external drivers.

1. Power Pins

3.3V Pin

- Provides a stable 3.3V output from the onboard regulator.
- Can power external sensors that require 3.3V.
- Should not be used to power high-current devices.

5V Pin (VIN / V5)

- Used to power the ESP32 board through USB or external 5V supply.
- Can also be used to power components that need 5V.

GND (Ground)

- All circuits must be connected to ground.
- Completes the electrical circuit for sensors, modules, and the ESP32.

2. Digital Input/Output Pins (GPIO)

ESP32 has 30+ GPIO pins, but not all are safe for general use. GPIO pins can read digital signals (0 or 1) and can output HIGH or LOW signals.

Commonly Used GPIO Pins

- GPIO 2 → Built-in LED, general use
- GPIO 4 → Analog/digital, often used with sensors
- GPIO 5 → General I/O, PWM
- GPIO 12, 13, 14, 15 → SPI pins but usable as GPIO

- GPIO 16, 17 → UART/General use
- GPIO 18, 19 → SPI/General use
- GPIO 21, 22 → I2C SDA/SCL (default)
- GPIO 23 → General-purpose digital pin
- GPIO 25, 26, 27 → ADC/DAC capable pins
- GPIO 32, 33 → ADC pins with touch support

3. Analog Input Pins (ADC)

ESP32 has 18 ADC pins:
GPIO 32 – GPIO 39 and some lower pins also support ADC.

These pins can read varying voltages (0 to 3.3V) and convert them into digital values.

Common examples:

- GPIO 34 → ADC1_CH6 (analog sensing)
- GPIO 35 → ADC1_CH7
- GPIO 36 (VP) → ADC1_CH0
- GPIO 39 (VN) → ADC1_CH3

Used for:

- LDR
- Gas sensors
- Temperature sensors
- Analog feedback

4. DAC Pins (Digital-to-Analog Output)

ESP32 has two DAC pins:

- GPIO 25 → DAC1

- GPIO 26 → DAC2

They output analog voltage (0–3.3V).
Useful for:

- Audio signals
- Motor control
- Sensor simulation

2.4 Conclusion

The ESP32 is a powerful, flexible, and cost-effective microcontroller that is perfectly suited for IoT-based security projects. Its Wi-Fi capability allows seamless communication with cloud services like Telegram, while its support for sensors and camera modules makes it a strong platform for intrusion detection. Understanding its features is essential for building dependable and scalable smart security systems.

Chapter 3

Hardware Description

3.1 Introduction

The hardware components used in the Smart Home Shield system form the foundation for achieving reliable real-time threat detection and alert communication. Each component is selected based on its accuracy, responsiveness, and compatibility with the ESP32 microcontroller platform. The system integrates multiple sensing elements, including the PIR sensor for motion detection, the LDR–laser pair for intrusion line monitoring, and the ESP32-CAM module for capturing photographic evidence. These components work together to form a multi-layered security mechanism capable of detecting different types of intrusions. Additional elements such as the buzzer provide immediate on-site alerts, while a regulated power supply ensures stable operation under continuous monitoring conditions. The keypad allows the user to arm and disarm the system securely. By combining these modules, the hardware architecture delivers a robust, low-cost, and scalable solution suitable for home and office security applications. This section provides detailed explanations of each hardware element and its role in the system.

3.2 Block Diagram

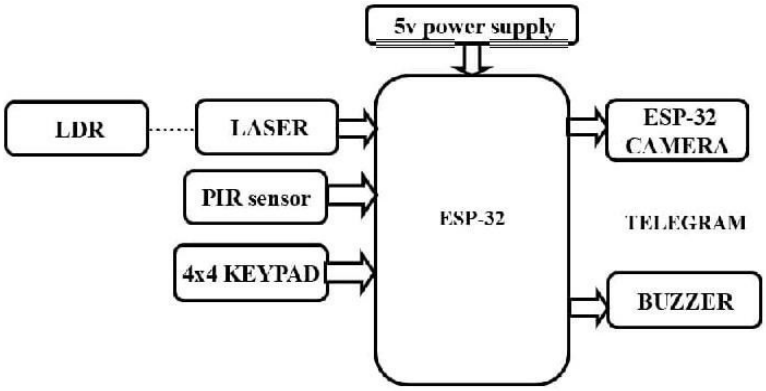

Fig 3.1: Block Diagram of the Project

3.3 ESP32 Camera Module

Fig 3.2: ESP 32 CAM

The ESP32-CAM module is one of the core components of the Smart Home Shield system, responsible for capturing images during intrusion events. It is a compact and cost-effective development board that integrates an ESP32-S microcontroller with an OV2640 camera module. The ESP32-CAM supports Wi-Fi connectivity, enabling it to immediately send captured images to the cloud or a mobile application such as Telegram. With its built-in antenna, microSD card slot, and onboard flash, the module is well-suited for real-time surveillance and image-processing applications.

The OV2640 camera used in the ESP32-CAM provides resolutions up to 2 megapixels, which is sufficient for indoor and outdoor security monitoring. The module supports JPEG image compression, allowing faster transmission over Wi-Fi with minimal bandwidth consumption. It operates at low power, making it ideal for continuous monitoring applications where energy efficiency is essential.

In the Smart Home Shield system, the ESP32-CAM is triggered whenever the PIR sensor or the laser–LDR mechanism detects suspicious activity. Upon receiving a trigger signal, the module captures an image and transmits it instantly to the user's Telegram account using the configured bot API. This ensures real-time visual verification of potential threats. The ESP32-CAM offers an excellent balance of performance, ease of use, and affordability, making it a perfect choice for IoT-based security applications.

3.3.1 Features of ESP32-CAM

- ESP32-CAM:
- The smallest 802.11b/g/n Wi-Fi BT SoC module.
- Low power 32-bit CPU, can also serve the application processor.
- Up to 160MHz clock speed, summary computing power up to 600 DMIPS.
- Built-in 520 KB SRAM, external 4MPSRAM.
- Embedded Lwin and Free RTOS.
- Supports STA/AP/STA+AP operation mode.
- Support Smart Config/Arikis technology.
- Support for serial port local and remote firmware upgrades (FOTA).

3.3.2 Description of ESP32-CAM

The ESP32-CAM is a compact and versatile microcontroller module designed specifically for image capture and wireless transmission applications. At its core, it uses the ESP32-S chip, which integrates a powerful dual-core processor, built-in Wi-Fi, and Bluetooth capabilities. This enables the module to handle both image processing tasks and network communication efficiently. The attached OV2640 camera sensor provides resolutions ranging from low-quality CIF to high-quality UXGA, making it adaptable for various surveillance and monitoring scenarios

In operation, the ESP32-CAM connects to a Wi-Fi network and runs a firmware program that enables image capture, compression, and transfer. When integrated with the Smart Home Shield system, it performs the crucial function of capturing intruder images and transmitting them instantly to the user's Telegram account. Its combination of performance, connectivity, and affordability makes it highly suitable for IoT-based smart home security solutions

3.3.3 Interfacing of ESP32-CAM to ESP32

Interfacing the ESP32-CAM module with the ESP32 development board is an essential step for enabling image capture and data communication within the Smart Home Shield system. Although the ESP32-CAM contains an onboard ESP32-S chip, it typically requires an external ESP32 or USB-to-TTL programmer for uploading code and providing initial configuration. For communication between the two devices, the UART interface is commonly used, where the TX pin of the ESP32-CAM is connected to the RX pin of the ESP32 board, and the RX pin of the ESP32-CAM is connected to the TX pin of the ESP32.

In addition, the GND pins must be connected together to establish a common reference point, while a stable 5V power supply is provided to the ESP32-CAM's 5V pin to ensure reliable operation of the camera module. The GPIO0 pin of the ESP32-CAM is grounded during code upload to enable bootloader mode and later released for normal execution.

3.4 Laser Module

Fig 3.3: LASER

The laser module plays an important role in the Smart Home Shield system by acting as a precise detection line for identifying unauthorized entry. It emits a narrow, focused beam of light that travels in a straight line and is aimed directly at the LDR sensor. As long as the beam remains uninterrupted, the system considers the area secure. When an intruder crosses the line, the laser beam is blocked, causing an immediate change in the LDR's output. This interruption is detected by the ESP32, which then triggers the ESP32-CAM to capture an image and activate the buzzer for an alert. The laser module operates on low power, typically 5V, making it suitable for continuous operation. Its high accuracy, long-range visibility, and

stability under different lighting conditions make it highly effective for intrusion detection. The simplicity and reliability of the laser module make it an essential component in real-time security applications.

3.4.1 Features of Laser Module

- **Focused Red Laser Beam (650 nm):** Emits a sharp and coherent red light beam suitable for long-distance alignment.
- **Low Operating Voltage (3–5V):** Works efficiently with microcontrollers like ESP32 without requiring complex power regulation.
- **Low Power Consumption:** Consumes minimal current, enabling continuous 24/7 operation without heating issues.
- **Compact and Lightweight Design:** Small module size makes installation flexible in doorways, hallways, and windows.
- **Stable and Continuous Beam Output:** Ensures reliable performance and minimizes false detection due to beam variation.
- **Long Transmission Range:** Can project a visible beam over several meters, suitable for indoor and outdoor security.

3.4.2 Description

The laser module is a compact electronic device designed to emit a narrow and concentrated beam of visible light, typically red in color. In the Smart Home Shield system, it serves as the primary source for creating an intrusion detection line. When positioned directly in front of an LDR sensor, the continuous laser beam keeps the system in a stable state. The module is built using a semiconductor laser diode enclosed within a small brass or plastic housing that provides durability and stable beam output. It operates on a low DC voltage, making it compatible with microcontroller-based circuits. When an object or person interrupts the beam, the sudden drop in light intensity allows the ESP32 to detect the intrusion and trigger an alert

3.4.3 Interfacing of Laser Module to ESP32

Interfacing the laser module with the ESP32 is straightforward, as the module only requires a stable DC power supply for operation. The laser module's VCC pin is connected to the 5V output pin of the ESP32, while the GND pin is connected to the ESP32 ground to complete the power circuit. Since the laser functions continuously and does not require direct control from the microcontroller, no GPIO pin is typically needed for switching unless the design includes a MOSFET or transistor for automated control. The laser is positioned so that its beam directly falls on the LDR sensor. When an intruder blocks the beam, the LDR's resistance changes, and this variation is detected by the analog input pin of the ESP32. This interruption acts as a trigger for the system to activate the ESP32-CAM, capture an image, and send a Telegram alert. Proper alignment between the laser and LDR ensures accurate intrusion detection.

3.5 LDR Sensor

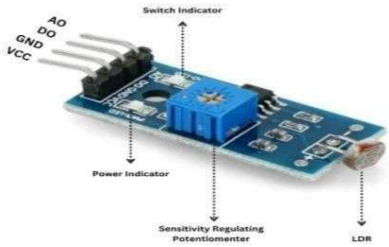

Fig 3.4: LDR Sensor

The Light Dependent Resistor (LDR) is an essential component of the Smart Home Shield system, used primarily for detecting interruptions in the laser beam. An LDR is a photo resistive device whose resistance varies based on the intensity of light falling on its surface. When the laser beam continuously hits the LDR, its resistance remains low, indicating a stable and secure state. However, when an object or intruder blocks the laser beam, the light intensity decreases sharply, causing the LDR's resistance to rise. This change is detected by the ESP32 through an analog input pin, triggering the intrusion detection process. The LDR is highly sensitive, inexpensive, and easy to interface, making it suitable for real-time security applications where constant monitoring of a detection line is required.

3.5.1 Features of LDR Sensor

- **Light-Sensitive Device:** The LDR's resistance decreases when exposed to light and increases in darkness.
- **Wide Range of Sensitivity:** Responds effectively to varying light intensities, including focused laser beams.
- **Low Cost and Easily Available:** Affordable and commonly used in simple electronic and IoT projects
- **Analog Output:** Provides resistance-based analog signals for precise detection.
- **Fast Response Time:** Quickly reacts to sudden changes in light, making it suitable for intrusion detection.
- **Low Power Consumption:** Operates efficiently without requiring high current or voltage.

3.5.2 Description (125 Words)

The Light Dependent Resistor (LDR) is a passive electronic component that changes its resistance based on the intensity of light falling on it. Made from semiconductor materials such as cadmium sulphide (CD's), the LDR exhibits low resistance in bright light and high resistance in darkness. This property makes it ideal for applications requiring light-based detection. In the Smart Home Shield system, the LDR is positioned to receive a focused laser beam continuously. Under normal conditions, the LDR maintains a steady, low-resistance state. When the laser beam is interrupted by an intruder, the resistance rises sharply, creating a detectable change in voltage at the ESP32's analog input pin. This simple yet highly effective principle allows the LDR to serve as a reliable intrusion sensor in real-time security setups.

3.5.3 Interfacing of LDR to ESP32

Interfacing the LDR sensor with the ESP32 is simple and requires only a few components. The LDR is typically connected in a voltage divider configuration using a fixed resistor. One end of the LDR is connected to the 3.3V supply, while the other end is connected to a resistor, which is then connected to ground. The junction points between the LDR and the resistor is connected to one of the analog input pins (ADC) of the ESP32. When the laser

beam falls on the LDR, its resistance decreases, producing a higher voltage at the ADC pin. When the beam is interrupted, the LDR's resistance increases, causing the voltage to drop. The ESP32 continuously reads these variations and identifies any disturbance in the light intensity. This change is used as a trigger to activate the ESP32-CAM, capture an image, and send a real-time alert to the user. Proper placement and alignment with the laser ensure accurate detection performance.

3.6 PIR Sensor

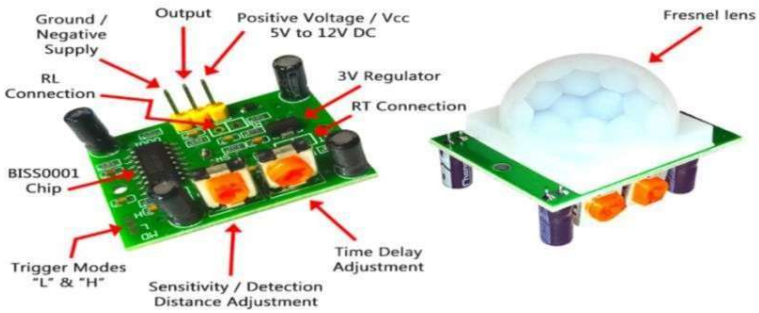

Fig 3.5: PIR Sensor

The Passive Infrared (PIR) sensor is a key component in the Smart Home Shield system, used for detecting human motion based on infrared radiation. Every living body emits heat in the form of infrared waves. The PIR sensor contains a pair of pyroelectric elements that detect changes in infrared levels within their sensing area. When a human enters its range, the sensor senses a sudden change in IR radiation and outputs a digital signal indicating motion. This makes the PIR sensor highly effective for intrusion detection, especially in indoor environments. It offers wide coverage, low power consumption, and fast response time. In the Smart Home Shield system, the PIR sensor acts as a primary motion detector to activate the ESP32-CAM, enabling the system to capture images and send alerts to the user instantly. Its reliability and simplicity make it an essential part of modern security systems.

3.6.1 Features of PIR Sensor

- **Passive Operation:** Detects infrared radiation without emitting any signals, making it energy efficient.
- **Human Motion Detection:** Specifically designed to sense human body heat, ensuring accurate intrusion detection.
- **Wide Detection Range:** Typically covers 5–7 meters with a detection angle of around 120°.
- **Low Power Consumption:** Requires very little current, ideal for continuous monitoring systems.
- **Fast Response Time:** Quickly detects motion and triggers the ESP32 without delay.
- **Digital Output:** Provides a simple HIGH/LOW signal, making interfacing with microcontrollers easy.

3.6.2 Description of PIR Sensor

The Passive Infrared (PIR) sensor is an electronic sensor designed to detect motion by measuring changes in infrared radiation emitted by objects, particularly humans. It contains two pyroelectric sensing elements placed side by side inside a protective Fresnel lens dome. These elements detect infrared energy and generate small electrical signals when there is a variation in the IR levels within their field of view. When a human enters or moves across the detection area, the difference in heat causes the pyroelectric elements to produce a voltage change. This signal is processed by the internal circuitry and converted into a digital output, typically HIGH when motion is detected and LOW when the area is stable. The PIR sensor is widely used due to its accuracy, low energy consumption, and ability to detect movement without physical contact. In the Smart Home Shield system, it serves as the first line of motion-based intrusion detection.

3.6.3 Interfacing of PIR Sensor to ESP32

Interfacing the PIR sensor with the ESP32 is simple because the sensor provides a digital HIGH or LOW output signal. The PIR module typically has three pins: VCC, GND, and OUT. The VCC pin is connected to the 5V or 3.3V supply of the ESP32, depending on the PIR module specifications, while the GND pin is connected to the ESP32 ground. The OUT

pin is connected to any digital GPIO pin of the ESP32 for reading the motion detection signal. When the PIR sensor detects movement, it sends a HIGH signal to the GPIO pin, which the ESP32 interprets as an intrusion event. This triggers the ESP32-CAM to capture an image and activate the buzzer for an alert. Sensitivity and delay adjustments on the PIR module allow fine-tuning of detection behaviour. Proper alignment and stable mounting ensure accurate and reliable motion detection within the monitored area.

3.7 Buzzer

Fig 3.6: Buzzer

The buzzer is a simple yet important component in the Smart Home Shield system, used to provide an immediate audible alert during intrusion events. It acts as a warning device that notifies nearby individuals of unauthorized activity. The system generally uses either an active buzzer or a passive buzzer. An active buzzer produces sound when power is applied, while a passive buzzer requires a PWM signal from the microcontroller. In this project, an active buzzer is commonly preferred due to its ease of interfacing. The buzzer is connected to one of the ESP32's digital GPIO pins, allowing the system to activate it whenever a sensor detects suspicious movement. When triggered, it emits a loud beep or continuous alarm sound, depending on the programmed output pattern. This sound alert adds another layer of security by discouraging intruders and informing users or neighbours of possible intrusion. Its low cost, compact size, and reliability make it a valuable part of the overall system.

3.7.1 Features of Buzzer

- **Simple Operation:** Produces sound immediately when voltage is applied, especially in active buzzers.
- **Low Power Consumption:** Requires minimal current, making it suitable for continuous or repeated alerts.
- **Compact Size:** Small and lightweight, allowing easy integration into compact security systems.
- **Loud Audible Output:** Generates a high-pitched sound that effectively alerts nearby individuals.
- **Wide Operating Voltage Range:** Many buzzers operate between 3V and 12V, providing flexibility in design.
- **Reliable Performance:** Offers stable sound output with long operational life.

3.7.2 Description

A buzzer is an audio signalling device that produces sound through mechanical, electromagnetic, or piezoelectric mechanisms. In electronic security systems like the Smart Home Shield, buzzers are commonly used to generate audible alerts whenever an intrusion is detected. The most widely used type is the piezoelectric buzzer, which creates sound when an electric signal causes a piezoelectric material to vibrate. Buzzers are available in two types: active buzzers, which contain an internal oscillator and require only a DC voltage to produce sound, and passive buzzers, which rely on an external driving signal such as PWM from a microcontroller. In this project, an active buzzer is preferred due to its simplicity and immediate response. When triggered by the ESP32, the buzzer emits a loud tone, effectively warning users and deterring intruders. Its compact size, low power consumption, and dependable operation make it an essential alert component in the system.

3.7.3 Interfacing of Buzzer to ESP32

Interfacing a buzzer with the ESP32 is simple and requires only a few connections. An active buzzer, which is commonly used in this project, needs just two pins: one for power and one for ground. The positive terminal of the buzzer is connected to any digital GPIO pin of the ESP32, while the negative terminal is connected to GND. The ESP32 controls the buzzer

by sending a HIGH signal to the GPIO pin, which activates the internal oscillator of the buzzer and produces sound. When the pin is set LOW, the buzzer turns off. This allows the system to generate different alert patterns as programmed. For louder sound output or higher voltage buzzers, a transistor can be used as a driver circuit. During an intrusion event, the ESP32 immediately triggers the buzzer to produce an audible alarm, warning users and deterring unauthorized access. This makes the buzzer a vital component for real-time security response.

3.8 Regulated Power Supply

The regulated power supply is a crucial component of the Smart Home Shield system, responsible for providing a stable and reliable voltage to all hardware modules. Devices like the ESP32, ESP32-CAM, PIR sensor, laser module, LDR setup, and buzzer require consistent DC voltage for proper operation. A regulated power supply ensures that these components receive a constant 5V or 3.3V, protecting them from voltage fluctuations that could cause malfunction or damage. It typically consists of a transformer, rectifier, filter capacitor, and voltage regulator IC such as the 7805. The regulator maintains a steady output even when the input supply varies. In IoT systems, stability is especially important because communication modules like the ESP32-CAM draw sudden bursts of current during Wi-Fi transmission. A well-regulated power source enhances reliability, prevents resets, and ensures uninterrupted functioning of the entire security system

3.8.1 Features of Regulated Power Supply

- **Stable Output Voltage:** Provides a constant 5V or 3.3V regardless of input fluctuations.
- **Overload Protection:** Many regulators include built-in protection against excess current.
- **Low Ripple Voltage:** Ensures smooth DC output with minimal noise, essential for microcontroller circuits.
- **High Efficiency:** Converts AC or varying DC input into clean, usable power with good efficiency.

- **Thermal Protection:** Prevents overheating by shutting down the regulator when temperature is high.

3.8.2 Description

A regulated power supply is an essential electronic unit designed to deliver a constant and stable DC voltage to different components of an electronic system. It internally uses key stages such as a rectifier to convert AC to DC, filter capacitors to reduce ripple, and a voltage regulator IC (like 7805 or AMS1117) to maintain a fixed output voltage. In the Smart Home Shield system, components such as the ESP32-CAM, ESP32 board, PIR sensor, LDR circuit, laser module, and buzzer rely on a precise 5V or 3.3V supply for proper operation. Voltage fluctuations can cause microcontroller resets, unstable Wi-Fi communication, or inaccurate sensor readings. By providing clean and regulated power, the supply ensures continuous and reliable functioning of the system. Its consistent voltage output enhances the overall stability and prevents damage caused by overvoltage or noise in the power lines, making it a critical part of IoT-based security setups.

3.8.3 Interfacing of Regulated Power Supply to ESP32

Interfacing a regulated power supply with the ESP32 is a straightforward process but essential for ensuring stable and reliable device operation. The ESP32 requires a steady 3.3V input, while modules like the ESP32-CAM typically operate on 5V. The regulated power supply provides these stable voltage levels, preventing fluctuations that could cause resets or malfunction. The output of the 5V regulator is directly connected to the 5V pin of the ESP32 or ESP32-CAM, while the 3.3V regulator (if separate) powers components requiring lower voltage. The ground (GND) of the regulated supply must be connected to the ESP32's GND to establish a common reference. This ensures clean and noise-free power delivery. When powering multiple sensors and modules, the regulated supply distributes current evenly without voltage drops. Proper interfacing guarantees uninterrupted Wi-Fi communication, accurate sensor readings, and stable performance of the entire Smart Home Shield system.

3.9 Keypad

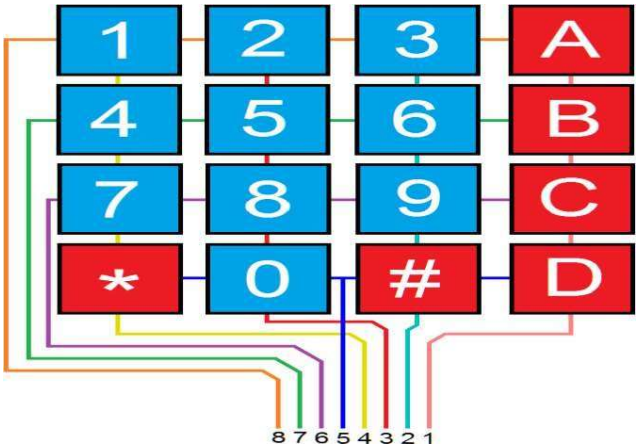

Fig 3.7: KEYPAD

The keypad is an important input device in the Smart Home Shield system, used primarily for arming and disarming the security mechanism. A typical 4×3 or 4×4 matrix keypad consists of a set of keys arranged in rows and columns. When a key is pressed, it creates a connection between a specific row and column, allowing the microcontroller to identify the pressed key. The keypad provides users with a simple and secure way to enter passcodes, ensuring that only authorized individuals can enable or disable the system. It is lightweight, durable, and easy to interface with microcontrollers like the ESP32. In the Smart Home Shield system, the keypad adds an additional authentication layer, preventing unauthorized persons from tampering with the system. Its low cost, simple wiring, and reliable performance make it a practical and effective input device for home security applications.

3.9.1 Features of Keypad

- **Matrix Layout:** Organized in rows and columns, enabling multiple key inputs using fewer GPIO pins.
- **Easy User Input:** Allows secure entry of passwords or commands directly into the system.
- **Low Power Consumption:** Consumes minimal energy, making it ideal for continuous use.
- **Durable Construction:** Designed to withstand repeated pressing without mechanical failure.

- **Simple Interfacing:** Works seamlessly with microcontrollers like ESP32 using digital I/O pins.

3.9.2 Description

A keypad is a widely used input device in embedded systems, allowing users to enter numeric or alphanumeric data through a series of buttons. In most electronic applications, a **matrix keypad** is preferred because it efficiently reduces the number of required input lines by arranging keys in rows and columns. When a key is pressed, it completes the circuit between a specific row and column, enabling the microcontroller to identify which key was activated. The keypad used in the Smart Home Shield system is typically a **4×3 or 4×4 membrane keypad**, known for its durability, lightweight design, and ease of installation. It provides a simple and secure method for entering passcodes to arm or disarm the system. The tactile response and reliable contact mechanism ensure accurate key detection. This makes the keypad an effective user authentication tool in home security applications, enhancing system safety and preventing unauthorized access.

3.9.3 Interfacing of Keypad to ESP32

Interfacing a matrix keypad with the ESP32 involves connecting the keypad's row and column pins to the digital GPIO pins of the microcontroller. A 4×4 keypad has eight pins—four for rows and four for columns—while a 4×3 keypad has seven pins. Each row and column wire is connected to a separate GPIO pin on the ESP32. The microcontroller scans the keypad by sending signals across the rows and reading the responses from the columns. When a key is pressed, the connection between a row and column is closed, enabling the ESP32 to detect the exact key through a scanning algorithm or keypad library. The keypad requires no additional power as it operates using the ESP32's I/O logic levels. In the Smart Home Shield system, the keypad is used to input passcodes for arming or disarming the system, providing an essential security layer. Proper interfacing ensures accurate key detection and smooth user interaction.

3.10 Conclusion

The hardware components discussed in this chapter collectively form the foundation of the Smart Home Shield system. Each module—ESP32-CAM, PIR sensor, LDR–laser setup, buzzer, keypad, and regulated power supply—plays a critical role in enabling real-time threat detection, system interaction, and reliable communication. The ESP32 and ESP32-CAM handle data processing, image capturing, and wireless transmission, while sensors such as the PIR and LDR provide accurate intrusion detection. Output devices like the buzzer ensure immediate on-site alerts, and the keypad allows secure user authentication. A stable power supply ties all components together, ensuring uninterrupted operation. By integrating these devices effectively, the system achieves a robust, efficient, and low-cost smart security solution suitable for modern homes and offices. This chapter highlights the importance of proper hardware selection and interfacing in building a dependable IoT-based security application.

Chapter 4

Software Tools

4.1 Introduction

Software tools play an essential role in the development and implementation of IoT-based systems like the Smart Home Shield. They help programmers write code, design circuits, configure communication platforms, and test the overall system operation. In this project, software tools were selected based on their flexibility, compatibility with ESP32 modules, and ease of use. Arduino IDE provides a stable environment for programming and uploading code. Express SCH helps create neat circuit diagrams used in documentation. Telegram Bot API ensures real-time image notifications to the user. Together, these tools enable efficient development, testing, and operation of the smart home security system.

Furthermore, software tools ensure that both hardware components and communication modules work together smoothly throughout the development cycle. They provide the necessary interface for configuring sensors, debugging errors, uploading test programs, and analysing output data. For IoT systems like the Smart Home Shield, software platforms also help manage Wi-Fi connectivity, device authentication, and message transfer to cloud services. The integration of multiple tools reduces development time and improves overall project accuracy. These tools also offer libraries and support documentation, which makes it easier for beginners and professionals to build reliable and scalable security solutions with enhanced functionality and performance.

4.2 Express SCH for Circuit Design

Express SCH is a simple yet effective circuit design software used to create professional schematic diagrams for electronic projects. In this project, Express SCH plays a crucial role in designing the complete wiring layout of the Smart Home Shield system. It allows easy placement of components such as ESP32, sensors, resistors, and power supply elements. Users can connect wires, label parts, and generate clear diagrams suitable for academic reports. The organized schematics help in understanding hardware connections, reducing wiring mistakes during implementation, and providing a visual representation of the system

architecture. Express SCH greatly improves documentation quality and enhances overall project clarity.

4.3 Arduino IDE

The Arduino Integrated Development Environment (IDE) is one of the primary software tools used in programming and developing the Smart Wearable Gesture Recognition Device. It provides a simple and user-friendly platform for writing, compiling, and uploading Embedded C/C++ code to the Arduino Nano. The IDE includes a built-in text editor, compiler, and serial monitor, allowing developers to test, debug, and monitor real-time data from sensors and communication modules.

Using the Arduino IDE, gesture-detection algorithms and control logic are implemented efficiently to manage flex sensors, the ADXL345 MEMS accelerometer, the GPS module, the GSM communication module, and the LCD display. The IDE also supports a wide range of libraries, making it easy to integrate hardware modules through I^2C, SPI, and UART interfaces. Its simplicity, cross-platform compatibility, and active community support make the Arduino IDE an essential tool for rapid development, testing, and deployment of embedded systems like the wearable gesture recognition device

4.4 Conclusion

The software tools used in the Smart Home Shield project play a crucial role in ensuring smooth development, accurate execution, and reliable performance of the entire IoT security system. Arduino IDE provides a simple yet powerful platform for writing and uploading the firmware required to control the ESP32 and ESP32-CAM modules. Its built-in debugging features, serial monitor, and wide library support significantly reduce development time and simplify complex tasks such as Wi-Fi configuration, sensor integration, and camera control. Express SCH contributes to creating professional and easily understandable circuit diagrams, which are essential for proper documentation and error-free hardware implementation.

Together, these software tools form a complete workflow that enhances productivity, ensures accurate monitoring, and delivers a reliable home security solution. Their combined

capability makes the Smart Home Shield system efficient, scalable, and user-friendly. Overall, the use of these tools demonstrates how software and hardware integration can create powerful IoT applications that improve safety, automation, and convenience in modern smart homes.

Chapter 5

Result

5.1 Introduction

This chapter presents the results and performance analysis of the developed system, Smart Home Shield: iot-Driven Real-Time Threat Protection System. The main objective of this project is to provide a reliable and efficient home security solution using multiple sensors such as PIR, LDR, and Laser sensors integrated with an iot-based monitoring system.

The system was designed and implemented to detect unauthorized access, motion, and abnormal environmental changes in real-time. Various test cases were conducted to evaluate the system's responsiveness, accuracy, and reliability under different conditions such as day/night environments and simulated intrusion scenarios.

The results obtained demonstrate how effectively the system detects threats and responds by triggering alarms and sending notifications to the user through the iot platform. The combination of multiple sensors ensures improved accuracy and reduces false alarms compared to single-sensor systems.

This chapter includes:

- Testing of individual sensors (PIR, LDR, Laser)
- Combined system performance
- Response time analysis
- Accuracy and reliability observations

Overall, the results validate that the proposed system successfully achieves real-time threat detection and provides an efficient smart home security solution.

 PIR Sensor: Successfully detected human motion within range (5–7 meters)

 LDR **Sensor:** Accurately detected light variations during day and night

 laser **sensor:** Effectively detected intrusion when beam was interrupted

5.2 Schematic Diagram

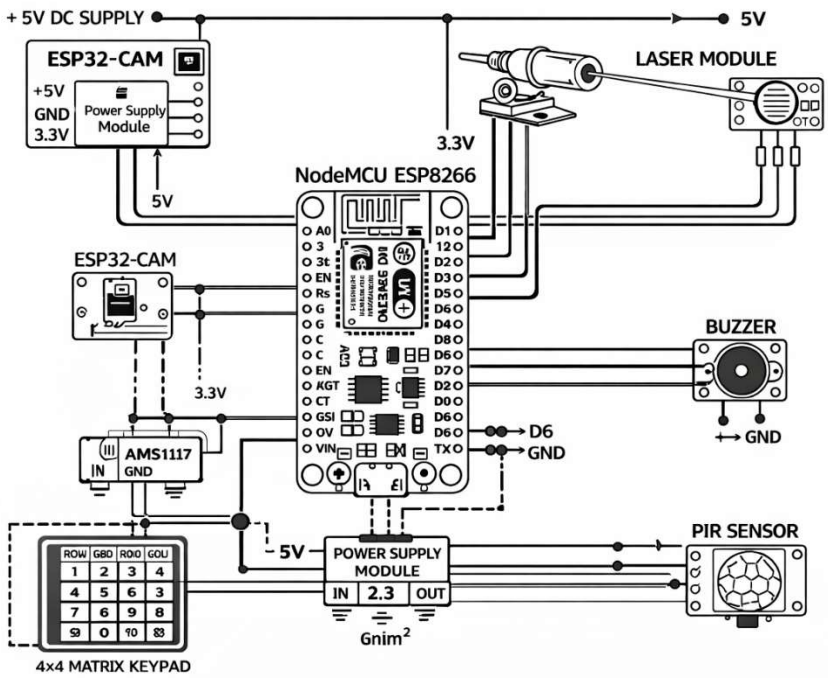

Fig 5.1: schematic diagram

The above schematic diagram represent the electrical connections of the proposed system. The nodemcu ESP8266 acts as the central controller interfaced with PIR sensor, LDR, laser module, keypad, and buzzer.

The sensors provide input signals, which are processed by the controller to detect threats. Upon detection, the buzzer is activated and alerts are sent through the iot platform.

5.3 Flowchart

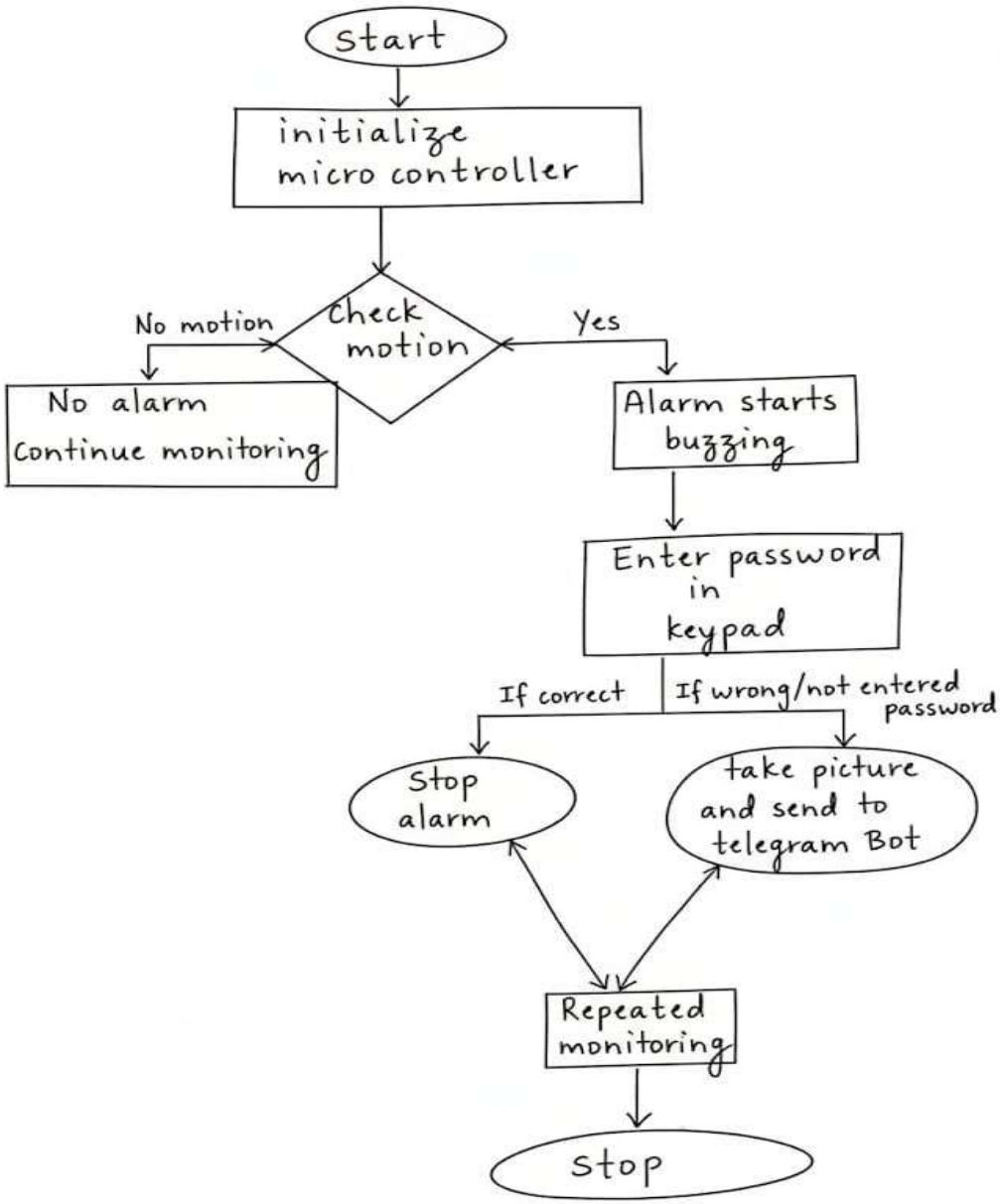

Fig 5.2: flowchart

The flowchart represents the working process of the Smart Home Shield system. Initially, the microcontroller is initialized and continuously monitors sensor inputs such as motion detection. If no motion is detected, the system remains in monitoring mode. When motion is detected, the alarm is activated, and the user is prompted to enter a password using the keypad. If the entered password is correct, the alarm is stopped. If the password is incorrect or not entered, the system captures an image using the ESP32-CAM and sends it to the user via a Telegram bot. The system then resumes monitoring, ensuring continuous security operation

5.4 Experimental setup

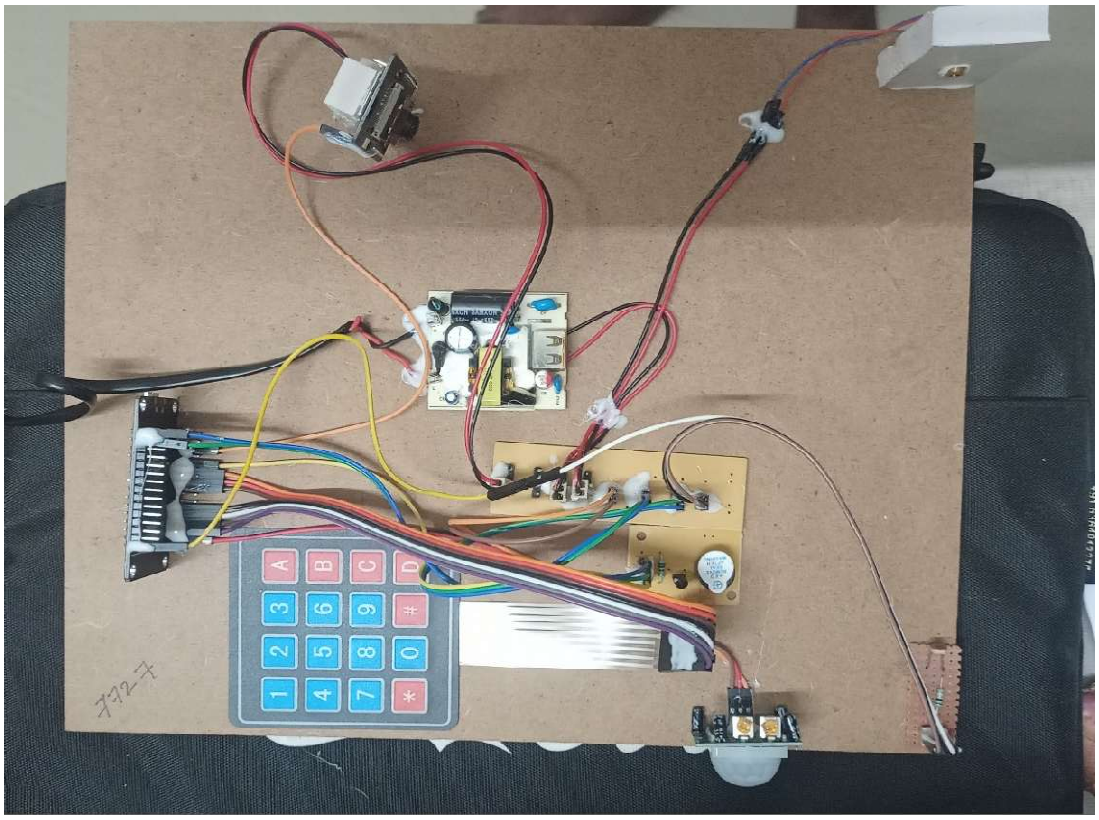

Fig 5.3: experimental setup

The experimental setup of the Smart Home Shield system consists of multiple hardware components integrated on a base platform to demonstrate real-time threat detection. The NodeMCU ESP8266 acts as the main controller, interfacing with input devices such as the PIR sensor for motion detection, an LDR with laser module for intrusion sensing, and a 4×4 matrix keypad for user authentication.

An ESP32-CAM module is also included to capture images when unauthorized access is detected. The system is powered using an AC to DC power supply module, providing regulated voltage to all components. A buzzer is connected as an alert mechanism to indicate security breaches. All components are interconnected using jumper wires and mounted on a wooden board for stability. The setup continuously monitors environmental conditions and responds instantly by triggering alarms and sending notifications, ensuring an effective and reliable smart security system.

5.5 Experimental results

STEP 1: Hardware setup: connected all the components according to circuit diagram

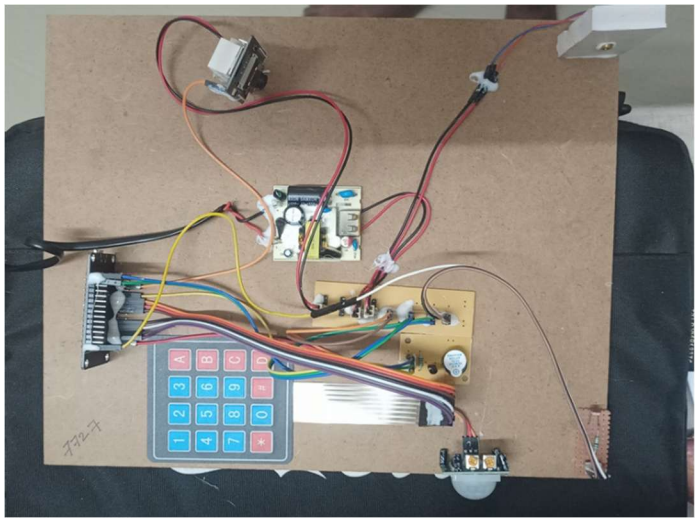

FIG 5.4: SETUP OF HARDWARE

STEP 2: Connect to network

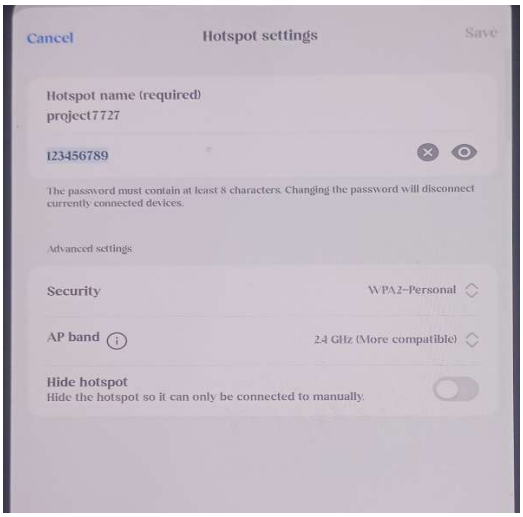

FIG 5.5: NETWORK

STEP 3: Given the required power supply to the circuit

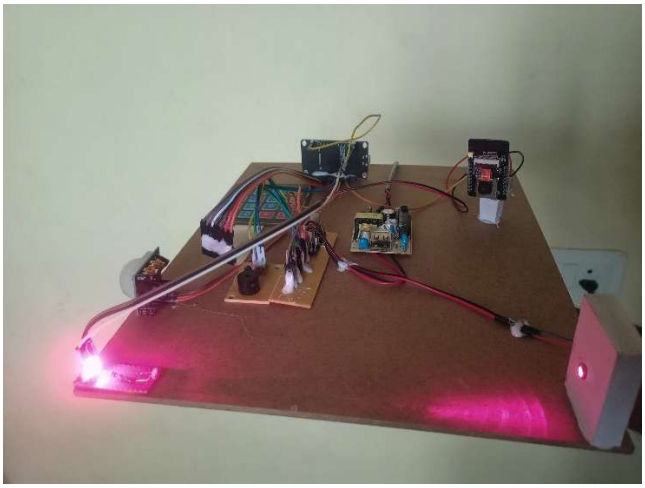

FIG 5.6: POWER SUPPLY

STEP 4: PIR and LDR sensors detect the motion and buzzer makes sound and if password enter correctly buzzer stop the sound while using the keypad

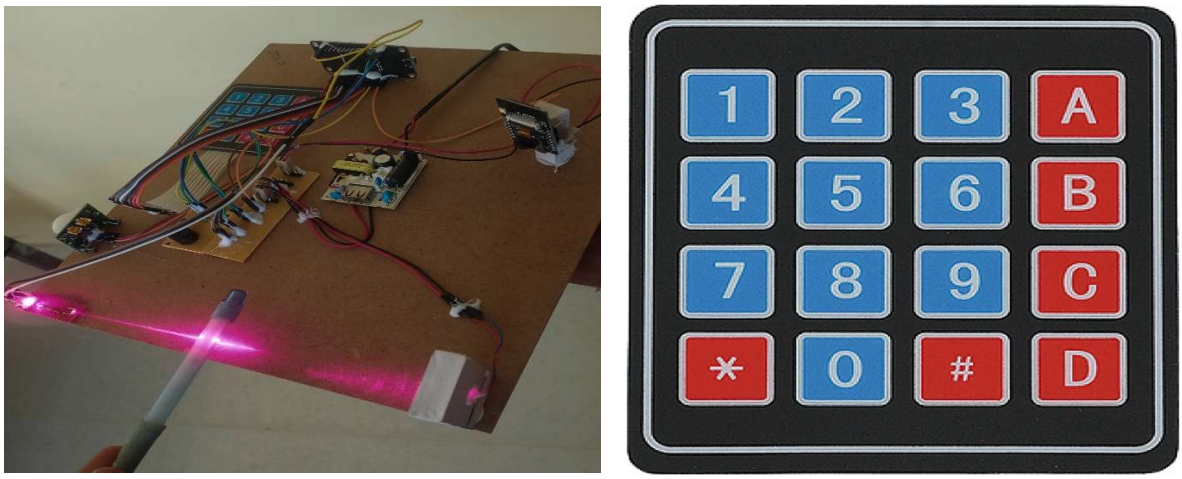

FIG 5.7: SENSORS & KEYPAD

STEP 5: If password not entered within 10seconds else password entered wrong. The Images taken by the ESP32 cam are sent to a telegram BOT

FIG 5.8: FINAL RESULT

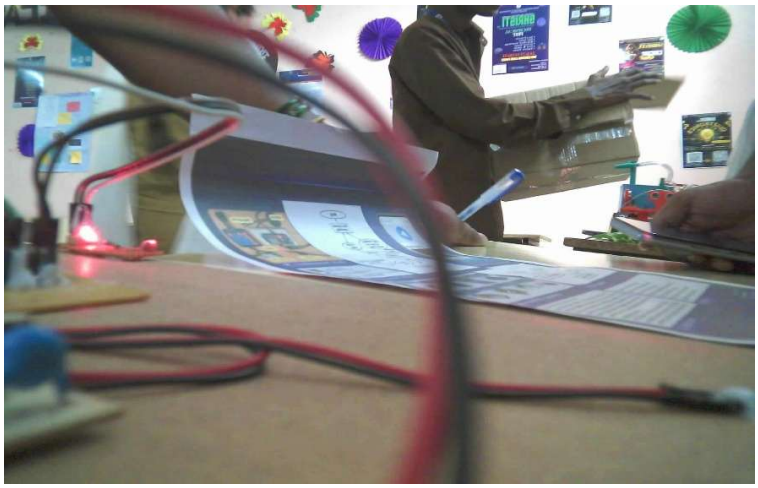

FIG 5.8: FINAL RESULT RECEIVED FROM TELIGRAM

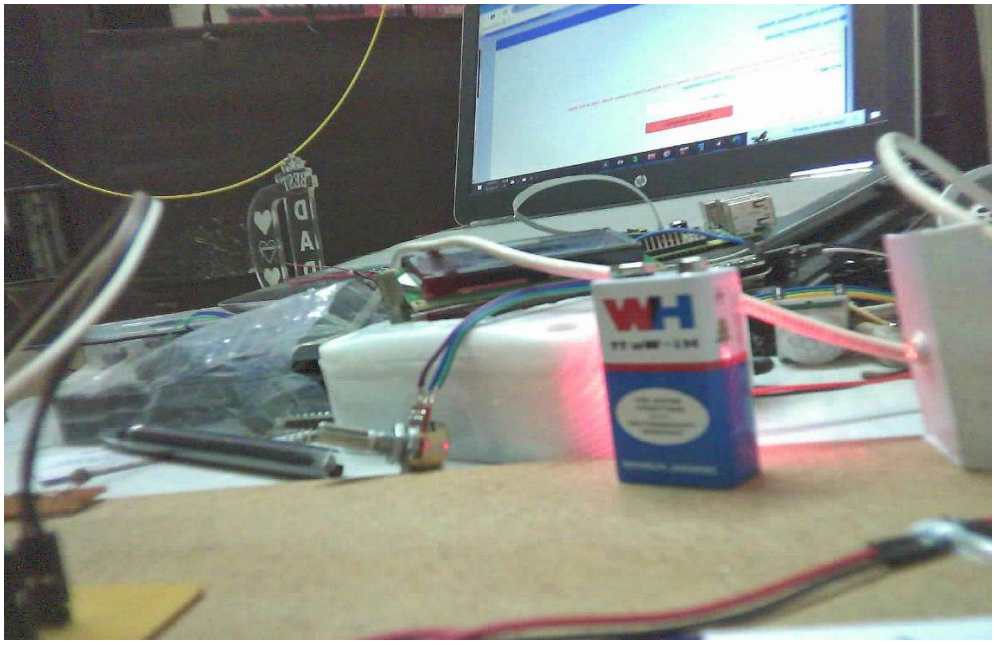

FIG 5.8: FINAL RESULT RECEIVED FROM TELIGRAM

5.6 Conclusions

The Smart Home Shield: iot-Driven Real-Time Threat Protection System has been successfully designed and implemented to enhance home security using modern technologies. The system integrates multiple sensors such as PIR for motion detection, LDR with laser for intrusion detection, and a keypad for user authentication, providing a multi-layered security approach. The nodemcu ESP8266 serves as the central controller, enabling real-time monitoring and communication through iot, while the ESP32-CAM enhances the system by capturing images during unauthorized access.

The experimental results demonstrate that the system is capable of detecting threats accurately and responding immediately by activating alarms and sending alerts to the user. The inclusion of password verification adds an extra layer of security, reducing false alarms. Additionally, the ability to capture and transmit images ensures better surveillance and evidence collection.
Overall, the project proves to be cost-effective, reliable, and scalable for real-world applications. It successfully achieves the objective of providing a smart, automated, and efficient home security solution. Future enhancements can further improve system performance by incorporating advanced technologies such as artificial intelligence and cloud-based analytics.

Chapter 6

Conclusions

The Smart Home Shield project successfully demonstrates how IoT technology can be used to create an efficient, low-cost, and intelligent home security system. By integrating the ESP32-CAM module with motion-detecting sensors such as PIR, LDR, and laser modules, the system is capable of identifying intrusions in real time and capturing images instantly. These images are then delivered to the user's Telegram application through an automated bot, enabling instant awareness even when the user is away from home. This immediate notification system greatly enhances safety compared to traditional alarm-based security setups.

The project also proves the effectiveness of wireless communication, automation, and cloud-based alerts in improving modern security solutions. The system is easy to install, scalable, and flexible enough to be extended for additional features such as video streaming, cloud storage, and face recognition. Although the project functions effectively, future improvements may include AI-based threat classification, use of GSM backup communication, and integration with smart home platforms like Google Home or Alexa. Overall, this work provides a reliable, practical, and user-friendly approach to home protection using IoT technology.

In addition to its core functions, the Smart Home Shield project highlights the importance of user-friendly design in modern security systems. The use of Telegram for sending images ensures that users receive alerts instantly without needing expensive monitoring equipment or complex applications. The modular structure of the system allows new sensors or automation features to be added easily, making it adaptable for various environments such as homes, offices, shops, and storage areas. The project also encourages awareness of IoT security challenges, such as data privacy and network reliability, which can be addressed in future versions. Overall, this system demonstrates how simple hardware and smart programming can create powerful real-time protection solutions.

References

1. **Patel, S., & Patel, A. (2020).** "Smart Home Security System Using IoT." *International Journal of Computer Applications*, 175(9), 1-5.
2. **Kumar, P., & Saini, R. (2019).** "IoT Based Home Security System with Smart Camera." *International Journal of Engineering and Technology*, 8(4), 12-18.